

DEGREE: PHYSICS

EXPERIMENTAL PROYECT II

EXPERIMENT 3: SOLAR CELLS

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1 Introduction

In this experiment we're going to learn how the solar cells work and the relation with the 'n' and 'p' type conductors. Also we're going to learn how a battery stores the energy of the solar cell and if the solar cell is connected to an electric motor how to change the movement if we polarize the cables.

2 Principles

The solar cells are capable of absorbing light and transforming a piece of this photon's energy into carriers of electrical current. We know this as electrons and holes.

This is an image of a cell structure:

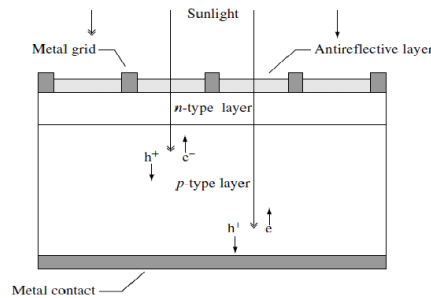


Figure 1: Cell structure

The function of the cell is that the sunlight hits from the top the front of the solar cell. Here the light can reach the semiconductor between the grid lines due to the metallic grid that forms one of the electrical contacts. Here it is absorbed and converted into electrical energy.

The facts that the light is transmitted to the semiconductor is because of the antireflective layer positioned between the grids. When we bring together a n-type and a p-type semiconductor to form a metallurgical junction, this creates a semiconductor diode.

This junction is established by employing a deposition process or through the diffusion or implantation of specific impurities (dopants). The diode's second electrical contact is established by a metallic layer situated on the back of the solar cell.

Fundamental properties of semiconductors

Using various semiconductors (silicon is the best choice) to manufacture solar cells, being silicon available in crystalline, polycrystalline, and amorphous forms. Additionally, this can be produced from alternative semiconductor materials like GaAs, GaInP, Cu(InGa)Se₂ and CdTe .

These solar cells have crystalline structures. Thus, atoms are organized in a periodic lattice (BCC and FCC among others). We have to describe the characteristic parameters of the lattice (Brillouin zone, length, occupation, etc.) to determine the properties.

We can T.I.S.E (time independent Schrödinger equation) to describe the dynamics of electrons forming semiconductors:

$$\nabla^2 \phi + \frac{2m}{\hbar} [E - U(\vec{r})] \phi = 0$$

Being m the electron mass, $\hbar$ the reduced Planck constant, E the energy of the electron, and U the periodic potential energy inside the semiconductor.

Electron-Hole conduction

We call the subatomic particles with a negative electric charge that orbit the nucleus of an atom in energy levels electrons. When we talk about conductive materials, these electrons can move freely and contribute to the electrical conduction.

On the other hand, the absence of the electron makes a hole in a valence band energy level; When this electron moves to a higher band, leaving the hole, its behaviour is like a positively charged particle.

The band theory explains how in solid materials the electrons fill energy levels, and in semiconductors the conduction of electricity occurs through electron movement in the conduction band or hole movement in the valence band. If there's an electric field, this causes electron movement in the opposite direction of the field, contributing to electron current, and similarly, holes can move in the direction of the electric field, contributing to the current too.

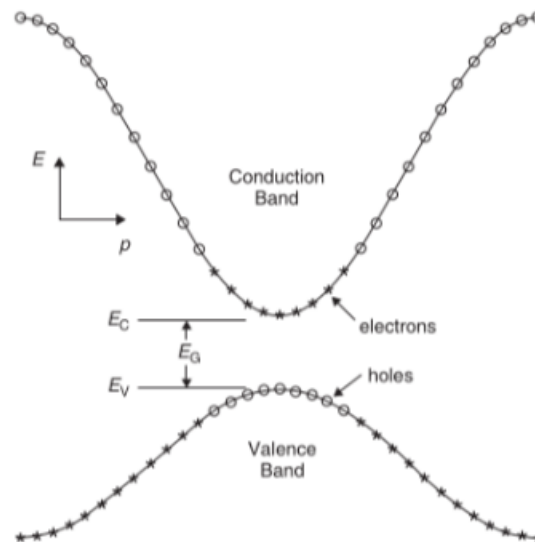


Figure 2: Energy band

3 Experimental Procedure

To carry out this experiment, we must first assemble it. The parts we will use are a solar cell, cables, a multimeter, an electric motor to detect if the cell is producing energy, a radiation lamp, an accumulator, and finally a support where everything we have said above will be mounted.

We did 3 different practices, so we will explain the 3 we have chosen and the assembly used.

3.1 First Experiment

In the first experiment, we wanted to know firstly if the solar cell is producing energy, so we had to connect the cables to the electric motor and the radiation lamp. As we can see in the photo:



Figure 3: Experiment 1 Assemble

Then, we wanted to know how the engine behavior changed if:

1. We change the polarity of the cables of the electric motor.
2. We change the intensity of the lamp from the regulator intensity.

3.2 Second Experiment

In the second experiment, we connect the solar cell to the lamp, the accumulator, the multimeter, and the electric motor. In this experiment, we're going to observe how the accumulator is like a battery and stores the energy from the solar cell, using a fully opaque black plate to cover the solar cell. And as before, we're going to observe the behavior if we change the polarity of the cables.

Here we have a photo of the assembly:



Figure 4: Experiment 2 Assemble

3.3 Third Experiment

Finally, for the last experiment, we have chosen the 4 one. In this one, we try fully opaque black plates larger and shorter to see how the solar cell catches the energy from the radiation lamp. We take the same

assembly as in experiment 2 but without the accumulator, so now we put the multimeter in volts (V) and take note of the data that we obtain.

Here we have a photo of the assembly:



Figure 5: Experiment 3 Assemble

4 Calculations

Now we are going to calculate and answer all the questions that were in each experiment.

4.1 First Experiment. The solar cell as an energy converter

1. What happens when we change the polarity of the cell connection cables?

If we change the polarity of the cables connected to the solar cell, we are exchanging the poles, so the electric motor rotates in the opposite direction.

2. On the brightness regulator, adjust different radiation intensities and observe the electric motor.

If we adjust the brightness to a greater or lesser degree, we can see that the motor reacts proportionally with the naked eye. That is, at a lower intensity, the motor spins more slowly, and at a higher intensity, it spins more quickly. This is due to the amount of radiation the solar cell receives and transforms into energy.

3. Indicate the energy conversions that take place in the solar cell and the electric motor.

A solar cell converts sunlight into electricity through the photovoltaic effect, where photons excite electrons in a semiconductor material (such as silicon), generating a direct current (DC). This electricity can be regulated using a DC-DC converter to adapt it to an electric motor.

If the motor is a DC motor, it can be connected directly; if it is an AC motor, an inverter is used for the conversion. When powering the motor, the current generates a magnetic field in its coils, causing the shaft to rotate and transforming electrical energy into mechanical motion. Since in this case we are dealing with direct current, we can connect it directly to the solar cell. For this reason, when we reverse the polarity of the wires, as we mentioned earlier, the direction of the motor's rotation changes.

4.2 Second Experiment. The solar cell as an energy converter or diode

1. In the beginning, the solar cell is operated without shading plates. What observation can be made?

With the wires of the instructions the experiment doesn't work like it should. The results might be that the motor turns to the right with 11mA in the current circuit. However, this occurs when we put the wires in the position of the photo bellow and with 8mA.



Figure 6: Experiment

2. Now, change the connections to the solar cell. What observation can be made?

When we change the wires position we observe that the direction of rotation change, rotation faster when it turns to the right.

3. Now, do again both experiments, but with the cover shade without irradiation. What observation can be made now?

When we cover the cells and repeat the first experiment our logic will tell us that wheel shouldn't turn. However, this still makes the same rotation but with a low velocity. This happens because the solar cell saves light when it has been a long time being illuminated.

The same happens with the second one. The change of rotation is produced put with a low velocity.

4.3 Third Experiment. The open circuit voltage of a solar cell / shading

1. What knowledge is obtained from measurement?

BLACK PLATE	0	50%	75%	100%
VOLTS	126	517	536	543

Figure 7: Data obtained from the measurement

As we can observe in the table, the percentages of the opaque black plate range from 0 to 100. This indicates the percentage that remains uncovered (not blocked by the plate) to the lamp's radiation.

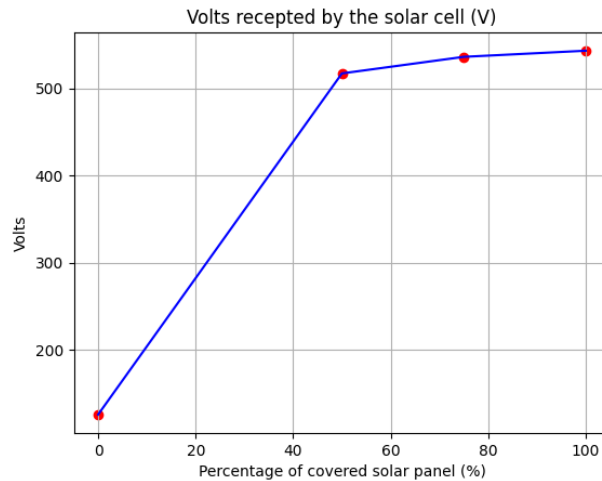


Figure 8: Volts / black plates graphic

As we can see in the table and later in the graph, the intensity increases exponentially as we gradually supply light radiation.

5 Discuss and evaluation

1. **What are the key steps in the manufacturing of solar cells used in this photovoltaic energy experiment?**

First, we treat the Silicon to be like a semiconductor, then we dop the material using phosphorus and boron to create p-type and n-type semiconductors that can generate electricity.

The surface of this is textured to reduce the reflection of the light and enhance absorption. To minimize more the reflection, it added a thin layer that is often of silicon nitride.

Then we print metal lines of Si or Al to collect and then transform electricity. Finally, the solar cells are en-capsuled between protective layers and they are tested to analyse their efficiency, durability and electrical performance.

2. **How does the efficiency of converting solar energy to electricity vary based on the intensity of incident light on the solar cells?**

The efficiency of this conversion of energy varies depending of the intensity of incident light. However, this doesn't occur of a lineal form. When the light increases more electron and holes are generated leading to higher output current. Although the voltage increases of a logarithmically form and limits the efficiency.

In conclusion, the efficiency improves with light up to a certain point.

3. **How does the ambient temperature affect the performance of solar cells, and what measures can be taken to optimize their efficiency under various conditions?**

Ambient temperature significantly impacts the performance of solar cells, primarily by affecting their efficiency and power output.

Ambient temperature affects solar cell performance by reducing efficiency and power output, mainly due to increased carrier recombination and decreased open-circuit voltage while excessive heat can degrade materials over time. Higher temperatures also increase electrical resistance, further lowering energy conversion rates.

To optimize performance, passive cooling methods like ventilated panel designs help dissipate heat, while active cooling systems (e.g., water or air-based) provide additional temperature control. Using low-temperature coefficient materials and heat-resistant coatings can further improve efficiency. Strategic panel orientation and tilt adjustments also aid in balancing light absorption and heat dissipation, ensuring better performance under varying conditions.

4. **What is a perovskite? What chemical structure does it have? Explain physically why it is such a promising material for photovoltaic cells.**

The Perovskite it is a mineral that belongs to group 4 according to the Strunz classification. It has the same structure as titanium trioxide, normally its structure is of the form:

We can see the structure in this picture:

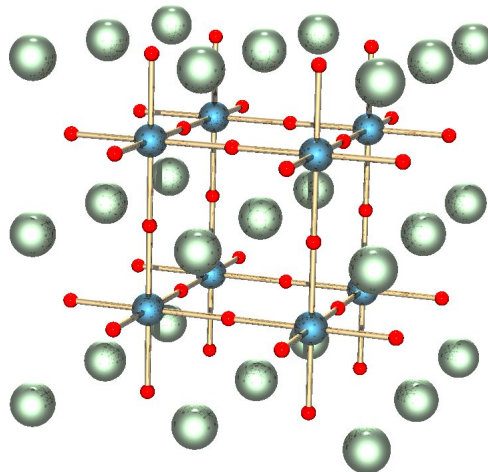
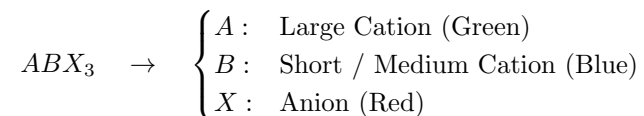


Figure 9: Perovskite structure



Perovskites are promising for solar cells due to their high light absorption, even in thin layers, and their tunable bandgap, which allows for efficiency optimization. They also have high charge carrier mobility and long diffusion lengths, improving electron collection and reducing recombination losses. Their low-cost fabrication is another advantage, as they can be processed at low temperatures using solution-based techniques, unlike silicon.

However, they face challenges such as instability to moisture, heat (as we said in the last question), and UV light, as well as environmental concerns due to the use of lead.

6 Bibliography

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